§ 1 Intro

§ 1.1 Classical Transport Theory

Drude - Boltzmann Method

① 碰撞间自由.

② 各向同性碰撞

③ 平均自由时 τ_c 与 x, v 相关

动量弛豫时 τ_m

④ 电子局域热平衡.

t 时刻, 动量 $p(t)$, e

$t \to t+dt$; f: 外力

$$p' = \begin{cases} 1-\frac{dt}{\tau_m} : f(t)\,dt \to p \\ \frac{dt}{\tau_m} : f(t)\,dt + p_{rand} \end{cases}$$ 碰撞后随机动量.

$$p(t+dt) = \left(1-\frac{dt}{\tau_m}\right)\left[p(t)+f(t)\,dt\right] + \left[p_{rand}+f(t)\,dt\right]\frac{dt}{\tau_m}$$

$$\bar{p}(t+dt) - \bar{p}(t) = -\frac{\bar{p}(t)\,dt}{\tau_m} + \bar{f}(t)\,dt + \frac{\bar{p}_{rand}\,dt}{\tau_m}, \quad \bar{p}_{rand}=0$$

$$\frac{d\bar{p}(t)}{dt} = -\frac{\bar{p}(t)}{\tau_m} + \bar{f}(t)$$

$f=0 \to p = p_0 e^{-t/\tau_m}$

$\alpha_m = \frac{\tau_c}{\tau_m}$, $\frac{1}{\tau_m} = \frac{\alpha_m}{\tau_c}$, 一次碰撞被阻动量的有效程度.

$l = v_d \tau_c$ 平均自由程

low α → smooth impurity / chiral state

$$\bar{p}(t) = \bar{p}_0 e^{-t/\tau_m} + \tau_m \bar{f}\left(1-e^{-t/\tau_m}\right)$$

电磁力 $\vec{f} = -e(\vec{E}+\vec{v}\times\vec{B})$

$$\dot{\vec{v}}_d = -\frac{\vec{v}_d}{\tau_m} - \frac{e}{m}(\vec{E}+\vec{v}_d\times\vec{B})$$

长时间稳态： $\vec{v}_d = 0$ ， $t \gg \tau_m$

零磁场： $\vec{B} = 0$ $\Rightarrow$ $\vec{v}_d = \frac{e\tau_m \vec{E}}{m} = \mu \vec{E}$

↳ Ohm定律：

$\vec{J} = \sigma_0 \vec{E}$ ， $\sigma_0 = e n_s \mu$

$= e n_s \mu \vec{E}$

$\mu_B = \frac{e\tau_m B}{m} = \omega_c \tau_m$

对二维材料： $\vec{E} \parallel x\text{-}y$ ， $\vec{B} \parallel \hat{z}$ 长时间稳态：

$$\vec{v}_d = -\frac{e\tau_m}{m}(\vec{E} + \vec{v}_d \times \vec{B})$$

$$\vec{E} = -\frac{m\vec{v}_d}{e\tau_m} - \vec{v}_d \times \vec{B}$$

$$= \begin{pmatrix} -\frac{m}{e\tau_m} & -B \\ B & \frac{m}{e\tau_m} \end{pmatrix} \begin{pmatrix} v_x \\ v_y \end{pmatrix} = \begin{pmatrix} \frac{1}{\mu} & -B \\ B & \frac{1}{\mu} \end{pmatrix} \vec{v}_d$$

$$= -\frac{1}{n_s e}\begin{pmatrix} \frac{1}{\mu} & -B \\ B & \frac{1}{\mu} \end{pmatrix} \vec{J} = \frac{1}{\sigma_0}\begin{pmatrix} 1 & \mu B \\ -\mu B & 1 \end{pmatrix} \vec{J} = \rho \vec{J}$$

对角电阻率 $\rho_{xx} = \rho_{yy} = \frac{1}{\sigma_0}$

Hall电阻率 $\rho_{xy} = -\rho_{yx} = \frac{\mu B}{\sigma_0}$

电导率： $\vec{J} = \sigma \vec{E}$ ， $\sigma = \rho^{-1}$

2D case： $\sigma_{xx} = \sigma_{yy} = \frac{\rho_{xx}}{\rho_{xx}^2 + \rho_{xy}^2} = \frac{\sigma_0}{1+\mu^2 B^2}$

$$\sigma_{xy} = \sigma_{yx} = -\frac{\rho_{xy}}{\rho_{xx}^2 + \rho_{xy}^2} = -\frac{\sigma_0 \mu B}{1+\mu^2 B^2}$$

$B \to 0$ 时： $\sigma_{xx} \to \rho_{xx}^{-1}$

当 $\vec{E} \parallel \hat{x}$

$$\begin{cases} J_x = \sigma_{xx} E_x = \frac{\sigma_0}{1+\mu^2 B^2} E_x \\ J_y = \sigma_{xy} E_x = \frac{\mu B \sigma_0}{1+\mu^2 B^2} E_x \end{cases}$$

ϕ ~ 两次散射间的转角

Hall angle： $\tan\phi = \frac{J_y}{J_x} = \mu B = \omega_c \tau_m$

Clean Limit: $\tau_m \to \infty$, $\mu \to \infty$ 没有散射、耗散.

$\rho_{xx} = 0$, $\sigma_{xx} = 0$ 只有Hall电流